ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q2, Q3, Q4

Question 2

(a) Let $x_1 = \theta$ and $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau + b\, x_2 - mg\ell \sin x_1]$.

Equilibrium: $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Because $-1 \le \sin\theta \le 1$ we need $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) Linearising: $A = [0,\ 1;\ \partial f_2/\partial x_1,\ \partial f_2/\partial x_2]$.

$\partial f_2/\partial x_1 = -(mg\ell/J)\cos\theta_e = -122.625 \times 0.8660 = -106.2$. $\partial f_2/\partial x_2 = -b/J = -2.5$.

$A = [0,\ 1;\ -106.2,\ -2.5]$, $B = [0;\ 1/J] =$ **$[0;\ 50]$**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss} = 2.4525 \sin\theta_{ss}$. At $\theta_d = 90°$ the right side $\neq 0$ while the left side $= 0$, so the joint cannot rest at θ_d.

With $\sin\theta_{ss} \approx 1$: $\theta_d - \theta_{ss} = 2.4525/50 = 0.00491$ rad = **0.281°**.

(d) New steady-state error = **0**, because the added term cancels the graviy torque exactly.

The poles do not change, because the added term is a feed-forward term that does not depend on the error, so it cannot affect the stabilty of the loop.

Question 3

(a) (1) Diferentiation amplifies noise in the encoder signal.

(2) Finite differencing needs a lot of memory and processing, which is slow on a micro-controler at 1 kHz.

(b) $\omega_c = 2\pi \times 20 = 125.66$ rad/s, $a = \omega_c T/2 = 0.0628$, Tustin transform.

$y_k =$ **$0.0591\,(u_k + u_{k-1}) + 0.8187\, y_{k-1}$**.

The DC gain is 1 since low-pass filters always have unity gain at DC.

(c) $A - LC = [-\ell_1,\ 1;\ -\ell_2,\ -2]$, so $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired: $(s + 10)(s + 10) = s^2 + 10s + 100$. Matching: $2 + \ell_1 = 10 \Rightarrow \ell_1 = 8$; $2(8) + \ell_2 = 100 \Rightarrow \ell_2 = 84$.

$L = [8,\ 84]^T$.

(d) Observer advantage: because it is a copy of the plant corrected by the encoder error, it gives a velocity estimate without the delay that a filter needs for noise rejection, and it can also estimate quantities no fliter on θ can, such as an unknown load torque.

A filter is the safer choice when the model is wrong or the true input is not known, e.g. unmodelled friction or backlash, a changing payload, or actuator saturation: the obsever would then be driven by a wrong u and drift, while the filter only uses measured data.

Question 4

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\, r$, i.e. $s^2 + 4s + 8 = 0$ so the poles are $s = -2 \pm 2j$.

DC gain $= bN_{ff}/(c + bk_1) = N_{ff}/8$. For unity gain $N_{ff} = 1/8 =$ **0.125**.

(b) With $b = 0.75$ ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

$\theta_{ss} =$ **0.8**. Tracking error $= (1 - 0.8)/1 \times 100 =$ **20%**. (Less than the 25% actuator loss, because the stiffness also dropped.)

(c) Char. poly: $s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 16s + 32$ (combining terms).

$2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$; $6 + k_1 = 16 \Rightarrow \mathbf{k_1 = 10}$; $K_I = \mathbf{32}$.

(d) For a constant r, in stedy state all derivatives vanish: $\dot{\xi} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stable so that a steady state exists.

Failure mode: integrator windup. When the actuator saturates the error persists, ξ keeps growing and causes a large overshoot afterwards. Remedy: anti-windup (clamp the integrator, conditional integration or back-calculation).